



C1

Your intuition is wrong

26 minutes

BY THE END OF THIS CAPSULE YOU WILL BE ABLE TO

Explain why you cannot predict what AI is good at by asking how hard the task looks.



Before I teach you anything,
let's find out what you already believe.

There are no marks for being right.



POLL 1 · THE FRONTIER CHECK

Which of these can today's best AI do reliably?

Select all you think it can.

Win a gold medal at the Maths Olympiad

Pass a professional law exam

Read an analog clock

Fold a pile of laundry

Predict a protein's 3D structure

Write a poem of exactly 50 words

Count the letters in a word

Drive a car through a city

Wooclap results

Answering counts towards your grades. Being right does not.



AT YOUR TABLE

3 min

Find one you disagreed on.

- 1 Compare answers with the people at your table.
- 2 Find one where you disagreed.
- 3 Argue it out. You have three minutes.

Introduce yourselves first if you haven't. You will be working with these people again.



So — which ones can it do?

ACES

Win a gold medal at the Maths Olympiad

Gold

Gemini Deep Think scored 35 points at the 2025 IMO — working end to end in natural language, inside the 4.5-hour limit.

FAILS

Read an analog clock

50.6%

The best model reads an analog clock correctly about half the time. Humans manage 90.1%.

Sources: AI Index 2026 · Mollick, Co-Intelligence · Abramson et al., Nature 630



So — which ones can it do?

ACES

Predict a protein's 3D structure

Solved

A 50-year open problem in biology. AlphaFold released predicted structures for essentially all ~200 million known proteins.

FAILS

Count the letters in a word

Unreliable

Ask how many r's are in “strawberry.” or s's in “Mississippi” or “Assassination”. It is not reading letters — it never sees them.

Sources: *AI Index 2026* · Mollick, *Co-Intelligence* · Abramson et al., *Nature* 630



So — which ones can it do?

ACES

Pass a professional law exam

Passes

GPT-4 passed the Uniform Bar Exam. Though how impressively is genuinely disputed — see the note below.

FAILS

Fold a pile of laundry

12%

Robots succeed at roughly 12% of real household tasks — against 89.4% in simulation.

Sources: *AI Index 2026* · Mollick, *Co-Intelligence* · Abramson et al., *Nature* 630



So — which ones can it do?

FAILS

Write a poem of exactly 50 words

Can't count

It will write you a genuinely good poem. It will not be 50 words. Mollick's own example.

IT DEPENDS

Drive a car through a city

Both

Waymo runs ~450,000 trips a week. Apollo Go completed 11 million driverless rides in China, up 175%. But only in mapped, geofenced cities.

Sources: *AI Index 2026* · Mollick, *Co-Intelligence* · Abramson et al., *Nature* 630



THE JAGGED FRONTIER

AI capability does not track how hard a task looks to a human.

Ethan Mollick, 2024

Coined the term in Co-Intelligence — your first pre-reading.

Stanford AI Index, 2026

Calls the same thing “jagged intelligence” — Takeaway #4, two years later.

Two independent sources. Same conclusion.



Why it is jagged

1 Architecture

It works in tokens, not letters — and in patches, not pixels. Some tasks are structurally impossible for it in a way that has nothing to do with difficulty.

We open this up in C3, after the break.

2 Unpredictability

Capability does not correlate with human-perceived difficulty. There is no rule you can learn. You have to test.

This is why A1 makes you compare two approaches empirically instead of arguing about them.

And the frontier moves with every model release. Any map you draw is already dated.



There is a pattern — just not the one you looked for

WHAT THE WINS SHARE

A verifiable objective

You can check an answer mechanically — which lets the system train against it and search at inference time.

Abundant curated data

Usually because an institution spent decades building it. The Protein Data Bank, not a web scrape.

A bounded output space

The set of acceptable answers is well defined.

WHAT THE FAILURES SHARE

Exact discrete state

Counting, length, position — held precisely across a process.

Continuous physical state

Deformable, partially observed, unrecoverable when wrong.

A dominant long tail

The rare cases are the ones that matter, and they are the thinnest in the data.

Notice what is absent from both lists: how hard the task looks to a human.



POLL 2 · YOUR VIEW

Which one surprised you most?

The clock

The 50-word poem

Counting letters

The law exam

Folding laundry

Driving a car

Wooclap results



You cannot reason your way to what AI can do.

You have to test it.

That is why A1 asks you to compare two approaches empirically — build both, measure both — rather than argue about which should be better.

Next: if you cannot predict it from difficulty — what CAN you predict it from?

Want the mechanisms? A four-page technical note on all eight tasks is on NTULearn.



APPENDIX

Two slides of mechanism,
for anyone who wants to know why.



Under the hood

Why each one lands where it does — mechanism, not anecdote.

Maths Olympiad

A proof is checkable. A verifiable reward permits reinforcement learning and search over many candidate reasoning traces at inference time — deliberation, not one forward pass.

Analog clock

Vision transformers embed image patches. Telling the time needs precise angular discrimination between two hands — exactly the fine geometric relation patch embedding blurs. There is no “measure this angle” primitive.

Protein structure

Multiple sequence alignment supplies co-evolutionary signal: residues that mutate together are likely in contact. Evolution ran the experiment; the PDB is decades of curated ground truth. AlphaFold 3 is diffusion-based.

Folding laundry

Cloth has unbounded degrees of freedom, self-occludes, and is partially observed. No simulator transfers. Demonstrations cannot be scraped — everyone is collected in real time by a person.

Driving

Operational design domain. Performance is bounded by the long tail, not the average — and rare events are both thinnest in the data and the ones that matter.

Counting letters

Byte-pair encoding. Deferred — we open this one up in C3, after the break.



How we know — and why to distrust these numbers

Every figure in this session came from a benchmark. Benchmarks are themselves contested.

ClockBench · RLBench · OSWorld

Each measures a narrow, defined slice. RLBench's 89.4% is rigid objects at known poses, in simulation. The 12% is a real house. Same field, incomparable conditions.

The bar exam claim was overstated

OpenAI reported ~90th percentile. Martínez (2024) found the comparison group was skewed toward repeat takers who had already failed. Against first-time takers: ~62nd. Against those who passed: ~48th overall, ~15th on essays.

Benchmarks contain errors

A review found invalid-question rates from 2% on MMLU Math to 42% on GSM8K. Frontier models gained 30 points in one year on a benchmark built to resist them — saturation, or contamination?

Sources: *AI Index 2026* · Martínez, E. (2024), *Artificial Intelligence and Law*

